

Class 10 Structural Bioinformatics 1

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PDB Statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178,795	21,825	12,773	343	226
2	Protein/Oligosaccharide	10,363	3,564	34	8	11
3	Protein/NA	9,106	6,335	287	24	7
4	Nucleic acid (only)	3,132	221	1,566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	214,078			
2	1	0	13,981			
3	0	0	15,759			
4	3	1	4,941			
5	0	0	237			
6	0	4	22			

```
sum(stats$Neutron)
```

```
[1] 88
```

The comma in these numbers leads to the numbers here being read as characters

```
c(100, "Barry")
```

```
[1] "100"    "Barry"
```

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
stats
```

```
# A tibble: 6 x 9
```

	`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1	Protein (only)	178795	21825	12773	343	226	84
2	Protein/Oligosacch~	10363	3564	34	8	11	1
3	Protein/NA	9106	6335	287	24	7	0
4	Nucleic acid (only)	3132	221	1566	3	15	3
5	Other	175	25	33	4	0	0
6	Oligosaccharide (o~	11	0	6	0	1	0

```
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.xray <- sum(stats$`X-ray`)
```

```
#n.em <-
```

```
n.total <- sum(stats$Total)
```

```
n.xray/n.total
```

```
[1] 0.8095077
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$EM)
n.total <- sum(stats$Total)

(n.xray/n.total) * 100
```

```
[1] 80.95077
```

```
(n.em/n.total) * 100
```

```
[1] 12.83843
```

Q2: What proportion of structures in the PDB are protein?

```
n.protein <- sum(stats$Total[grept("Protein", stats$`Molecular Type`)])
(n.protein / n.total) * 100
```

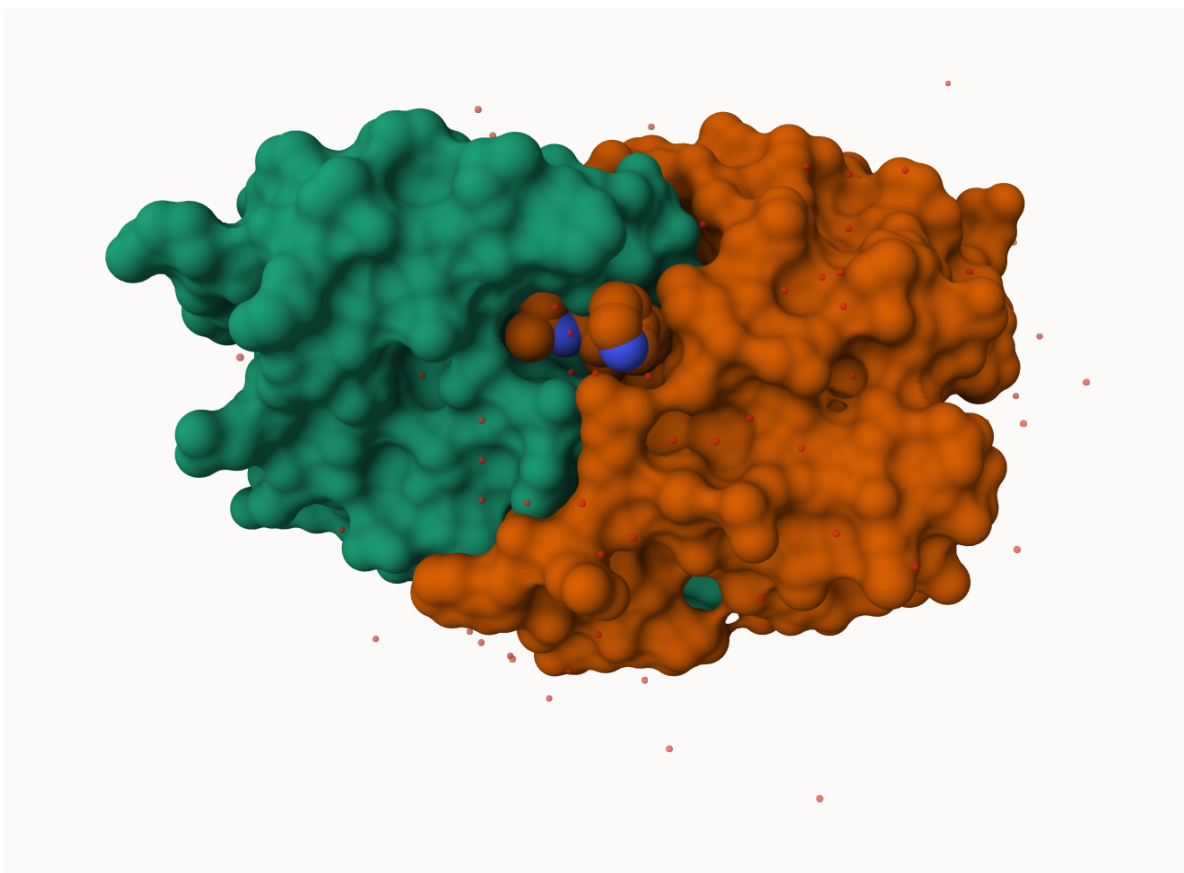
```
[1] 97.9118
```

Q3. Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Skip...

Visualizing the HIV-1 protease structure

We can use Molstar viewer online: <https://molstar.org/viewer/>



A clean image showing the catalytic ASP25 amino acids in both chains of the HIV-PR dimer along with the inhibitor and all important active site water



Bio3D package for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
head( pdb$atom )
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
library(bio3dview)
```

```
# view.pdb(pdb)
```

```
# Select the important ASP 25 residue
```

```
sele <- atom.select(pdb, resno=25)
```

```
# and highlight them in spacefill representation
```

```
# view.pdb(pdb, cols=c("steelblue","lightpink"),
```

```
      # highlight = sele,
```

```
      # highlight.style = "spacefill")
```

Predicting functional motions of a single structure

Read an ADK structure from the PDB database:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

```
Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

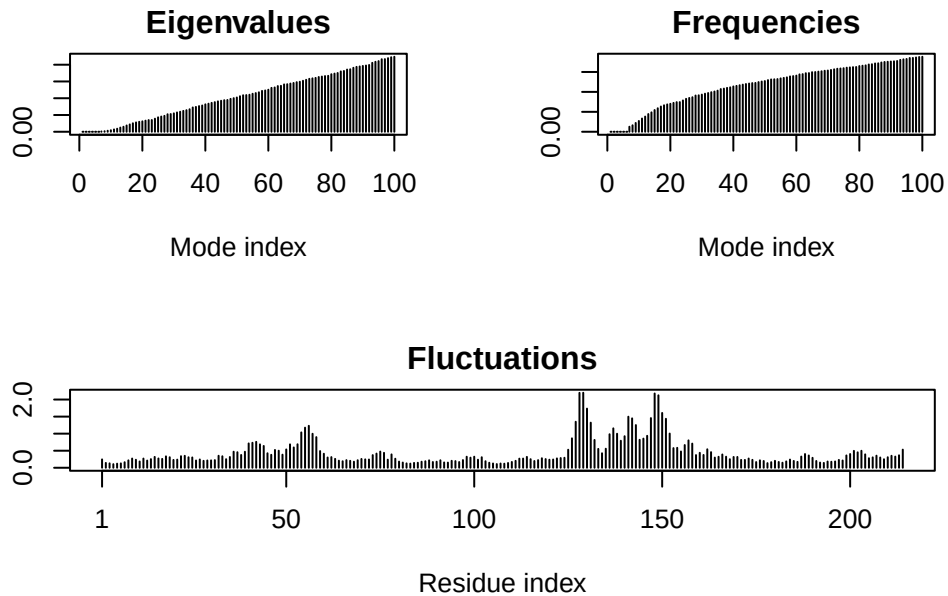
```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.034 seconds.
Diagonalizing Hessian... Done in 0.356 seconds.
```

```
plot(m)
```



Write out our results as a wee trajectory/movie of predicted motions:

```
mktrj(m, file="adk_m7.pdb")
```

Comparative Analysis with PCA

First step: Find an ADK sequence:

```
library(bio3d)
id <- "lake_A" ## Change this to run a different analysis
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

Next step, is search the PDB dtata base fore all related entries:

```
blast <- blast.pdb(aa)
```

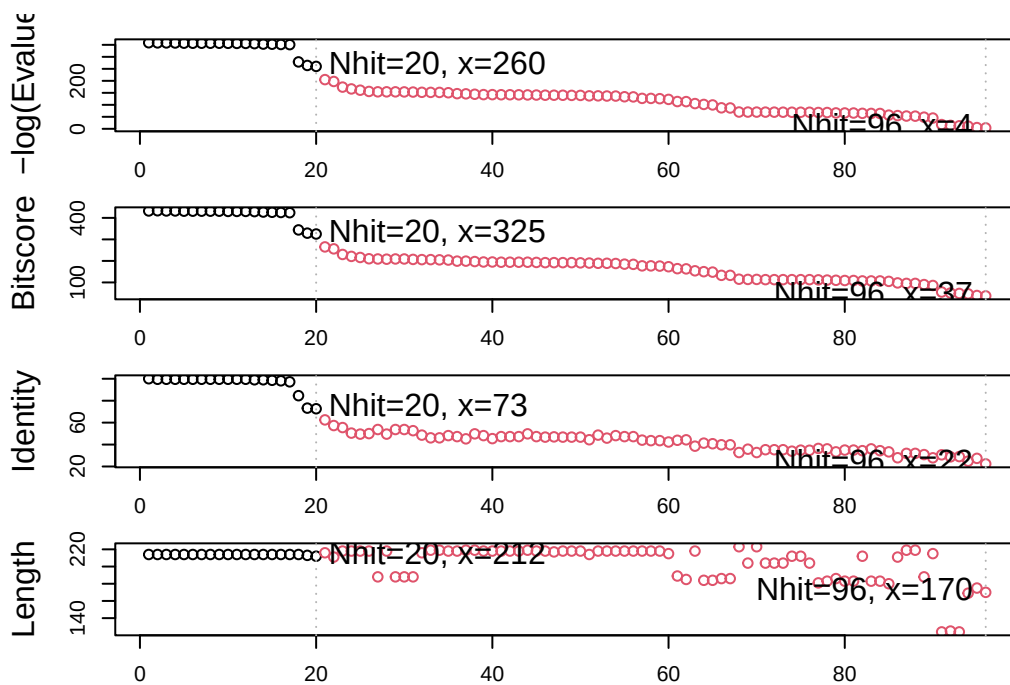

Searching ... please wait (updates every 5 seconds) RID = SR0UKSJT014

 Reporting 96 hits

```
hits <- plot(blast)
```

```
* Possible cutoff values: 260 3
  Yielding Nhits: 20 96
```

```
* Chosen cutoff value of: 260
  Yielding Nhits: 20
```



```
blast$hit.tbl
```

	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_847717	1AKE_A	100.000	214	0	0	1
2	Query_847717	8BQF_A	99.533	214	1	0	1
3	Query_847717	4X8M_A	99.533	214	1	0	1
4	Query_847717	6S36_A	99.533	214	1	0	1
5	Query_847717	9R6U_A	99.533	214	1	0	1
6	Query_847717	9R71_A	99.533	214	1	0	1

7	Query_847717	8Q2B_A	99.533	214	1	0	1
8	Query_847717	8RJ9_A	99.533	214	1	0	1
9	Query_847717	6RZE_A	99.533	214	1	0	1
10	Query_847717	4X8H_A	99.533	214	1	0	1
11	Query_847717	3HPR_A	99.533	214	1	0	1
12	Query_847717	1E4V_A	99.533	214	1	0	1
13	Query_847717	5EJE_A	99.065	214	2	0	1
14	Query_847717	1E4Y_A	99.065	214	2	0	1
15	Query_847717	3X2S_A	98.598	214	3	0	1
16	Query_847717	6HAP_A	98.131	214	4	0	1
17	Query_847717	6HAM_A	97.196	214	6	0	1
18	Query_847717	8PVW_A	84.579	214	6	2	1
19	Query_847717	4K46_A	73.239	213	57	0	1
20	Query_847717	4NP6_A	72.642	212	58	0	2
21	Query_847717	3GMT_A	62.500	216	75	1	2
22	Query_847717	4PZL_A	57.346	211	86	2	2
23	Query_847717	5G3Y_A	55.505	218	88	2	1
24	Query_847717	5G3Z_A	50.459	218	99	2	1
25	Query_847717	5G40_A	49.541	218	101	2	1
26	Query_847717	5X6J_A	50.000	218	98	3	1
27	Query_847717	2C9Y_A	53.723	188	83	1	1
28	Query_847717	1S3G_A	49.541	218	99	3	1
29	Query_847717	9GR0_A	53.723	188	83	1	1
30	Query_847717	9FL7_D	53.723	188	83	1	1
31	Query_847717	1AK2_A	52.660	188	85	1	1
32	Query_847717	3BE4_A	48.611	216	102	3	2
33	Query_847717	1AKY_A	46.119	219	108	3	1
34	Query_847717	3AKY_A	46.119	219	108	3	1
35	Query_847717	3FB4_A	48.165	218	104	2	1
36	Query_847717	4QBI_A	47.248	218	106	2	1
37	Query_847717	1DVR_A	45.205	219	110	3	1
38	Query_847717	3DKV_A	49.772	219	99	3	1
39	Query_847717	3DLO_A	48.165	218	104	2	1
40	Query_847717	1ZIN_A	45.413	218	110	2	1
41	Query_847717	2P3S_A	47.248	218	106	2	1
42	Query_847717	2EU8_A	47.248	218	106	2	1
43	Query_847717	1P3J_A	47.248	218	106	2	1
44	Query_847717	4QBF_A	49.772	219	99	3	1
45	Query_847717	2ORI_A	47.248	218	106	2	1
46	Query_847717	5X6I_A	46.789	218	107	2	1
47	Query_847717	2QAJ_A	47.005	217	106	2	1
48	Query_847717	2007_A	46.789	218	107	2	1
49	Query_847717	20SB_A	46.789	218	107	2	1

50	Query_847717	4MKF_A	46.789	218	107	2	1
51	Query_847717	3TLX_A	44.393	214	106	3	2
52	Query_847717	4MKH_A	48.624	218	101	3	1
53	Query_847717	4QBH_A	45.872	218	109	2	1
54	Query_847717	4TYQ_A	48.165	218	102	3	1
55	Query_847717	4QBG_B	47.248	218	104	3	1
56	Query_847717	4TYP_A	47.248	218	104	3	1
57	Query_847717	4JKY_A	44.037	218	103	5	1
58	Query_847717	2RGX_A	43.578	218	104	4	1
59	Query_847717	4JLO_A	43.578	218	104	5	1
60	Query_847717	1ZAK_A	42.326	215	112	3	1
61	Query_847717	1ZD8_A	43.915	189	96	3	1
62	Query_847717	2AK3_A	44.324	185	101	2	1
63	Query_847717	4NTZ_A	38.532	218	119	4	1
64	Query_847717	2AR7_A	41.304	184	102	3	1
65	Query_847717	3NDP_A	40.761	184	103	3	1
66	Query_847717	1P4S_A	39.785	186	77	2	1
67	Query_847717	2CDN_A	39.785	186	77	2	1
68	Query_847717	3LOP_A	32.735	223	131	7	1
69	Query_847717	5X6L_A	35.784	204	98	3	3
70	Query_847717	2XB4_A	32.735	223	131	7	1
71	Query_847717	5XRU_A	35.294	204	99	3	3
72	Query_847717	5YCC_A	35.294	204	99	3	3
73	Query_847717	5X6K_A	35.294	204	99	3	3
74	Query_847717	5XZ2_A	33.962	212	107	3	3
75	Query_847717	5YCF_A	34.906	212	105	3	3
76	Query_847717	5YCB_A	34.804	204	100	3	3
77	Query_847717	5YCD_A	36.464	181	87	2	3
78	Query_847717	3ADK_A	36.066	183	89	3	3
79	Query_847717	3UMF_A	33.333	186	92	3	3
80	Query_847717	1Z83_A	34.973	183	91	3	3
81	Query_847717	7X7S_A	34.973	183	91	3	3
82	Query_847717	3CM0_A	34.434	212	106	5	3
83	Query_847717	7DE3_A	36.066	183	89	5	3
84	Query_847717	8X1G_A	34.426	183	92	3	3
85	Query_847717	7N6G_6M	33.333	180	108	4	1
86	Query_847717	1UKY_A	27.962	211	117	5	3
87	Query_847717	1TEV_A	31.963	219	109	7	3
88	Query_847717	7E9V_A	31.963	219	109	7	3
89	Query_847717	2BWJ_A	30.851	188	98	4	3
90	Query_847717	1QF9_A	27.907	215	117	5	3
91	Query_847717	9FQR_Xf	30.645	124	83	2	1
93	Query_847717	9FQR_Xc	28.800	125	56	4	1

94	Query_847717	9D2F_D	29.032		124	56	4	1	
95	Query_847717	7N6G_6A	24.852		169	84	6	76	
96	Query_847717	9D2F_A	27.429		175	78	6	1	
92	Query_847717	9FQR_Xf	22.353		170	83	2	1	
	q.end	s.start	s.end	evalue	bitscore	positives	mlog.evalue	pdb.id	acc
1	214	1	214	1.78e-156	432.0	100.00	358.626661	1AKE_A	1AKE_A
2	214	21	234	2.91e-156	433.0	100.00	358.135121	8BQF_A	8BQF_A
3	214	1	214	3.18e-156	432.0	100.00	358.046393	4X8M_A	4X8M_A
4	214	1	214	4.67e-156	432.0	100.00	357.662115	6S36_A	6S36_A
5	214	1	214	1.04e-155	431.0	99.53	356.861469	9R6U_A	9R6U_A
6	214	1	214	1.23e-155	431.0	99.53	356.693675	9R71_A	9R71_A
7	214	1	214	1.24e-155	431.0	99.53	356.685578	8Q2B_A	8Q2B_A
8	214	1	214	1.24e-155	431.0	99.53	356.685578	8RJ9_A	8RJ9_A
9	214	1	214	1.34e-155	431.0	99.53	356.608020	6RZE_A	6RZE_A
10	214	1	214	1.76e-155	430.0	99.53	356.335376	4X8H_A	4X8H_A
11	214	1	214	2.50e-155	430.0	99.53	355.984399	3HPR_A	3HPR_A
12	214	1	214	2.64e-155	430.0	99.53	355.929910	1E4V_A	1E4V_A
13	214	1	214	7.91e-155	429.0	99.07	354.832562	5EJE_A	5EJE_A
14	214	1	214	4.58e-154	427.0	99.07	353.076405	1E4Y_A	1E4Y_A
15	214	1	214	7.67e-154	426.0	98.60	352.560788	3X2S_A	3X2S_A
16	214	1	214	2.27e-153	425.0	98.60	351.475739	6HAP_A	6HAP_A
17	214	1	214	4.64e-153	424.0	98.60	350.760805	6HAM_A	6HAM_A
18	214	1	187	8.46e-122	344.0	85.05	278.780032	8PVW_A	8PVW_A
19	213	1	213	2.06e-115	329.0	84.98	264.074580	4K46_A	4K46_A
20	213	5	216	1.14e-113	325.0	84.43	260.061087	4NP6_A	4NP6_A
21	211	10	225	8.94e-90	265.0	71.30	205.042123	3GMT_A	3GMT_A
22	209	26	235	2.13e-86	256.0	74.41	197.266196	4PZL_A	4PZL_A
23	214	1	213	3.16e-76	230.0	68.81	173.845895	5G3Y_A	5G3Y_A
24	214	1	213	6.04e-73	221.0	69.27	166.290308	5G3Z_A	5G3Z_A
25	214	1	213	1.24e-70	216.0	68.35	160.965845	5G40_A	5G40_A
26	213	1	212	2.26e-68	210.0	65.60	155.760422	5X6J_A	5X6J_A
27	184	17	204	1.14e-67	209.0	69.68	154.142173	2C9Y_A	2C9Y_A
28	213	1	212	1.17e-67	208.0	65.14	154.116197	1S3G_A	1S3G_A
29	184	16	203	1.24e-67	209.0	69.68	154.058090	9GR0_A	9GR0_A
30	184	22	209	1.66e-67	209.0	69.68	153.766384	9FL7_D	9FL7_D
31	184	17	204	2.94e-67	207.0	70.21	153.194792	1AK2_A	1AK2_A
32	213	7	217	7.91e-67	206.0	68.06	152.205073	3BE4_A	3BE4_A
33	214	5	218	1.12e-66	206.0	65.75	151.857287	1AKY_A	1AKY_A
34	214	5	218	1.36e-66	205.0	65.30	151.663131	3AKY_A	3AKY_A
35	214	1	213	5.00e-66	204.0	65.14	150.361178	3FB4_A	3FB4_A
36	214	1	213	2.42e-64	199.0	65.60	146.481678	4QBI_A	4QBI_A
37	214	5	218	5.22e-64	199.0	64.84	145.712949	1DVR_A	1DVR_A
38	214	1	213	2.81e-63	197.0	66.67	144.029676	3DKV_A	3DKV_A

39	214	1	213	1.24e-62	195.0	66.97	142.545164	3DLO_A	3DLO_A
40	214	1	213	1.38e-62	195.0	65.60	142.438192	1ZIN_A	1ZIN_A
41	214	1	213	2.08e-62	194.0	66.97	142.027908	2P3S_A	2P3S_A
42	214	1	213	2.18e-62	194.0	66.97	141.980951	2EU8_A	2EU8_A
43	214	1	213	2.95e-62	194.0	66.97	141.678471	1P3J_A	1P3J_A
44	214	1	213	4.71e-62	194.0	66.21	141.210588	4QBF_A	4QBF_A
45	214	1	213	7.38e-62	193.0	66.97	140.761502	2ORI_A	2ORI_A
46	214	1	213	1.40e-61	192.0	66.51	140.121218	5X6I_A	5X6I_A
47	213	1	212	1.51e-61	192.0	66.82	140.045581	2QAJ_A	2QAJ_A
48	214	1	213	1.76e-61	192.0	66.51	139.892377	2007_A	2007_A
49	214	1	213	3.01e-61	192.0	66.51	139.355751	20SB_A	20SB_A
50	214	1	213	3.35e-61	191.0	66.51	139.248730	4MKF_A	4MKF_A
51	211	31	235	2.11e-60	190.0	64.95	137.408418	3TLX_A	3TLX_A
52	213	3	214	2.16e-60	189.0	65.60	137.384997	4MKH_A	4MKH_A
53	214	1	213	4.32e-60	189.0	64.68	136.691850	4QBH_A	4QBH_A
54	213	1	212	5.32e-60	188.0	65.60	136.483632	4TYQ_A	4TYQ_A
55	213	1	212	9.31e-59	185.0	65.60	133.621431	4QBG_B	4QBG_B
56	213	1	212	1.75e-58	184.0	65.14	132.990320	4TYP_A	4TYP_A
57	214	1	203	8.03e-56	177.0	66.51	126.861581	4JKY_A	4JKY_A
58	214	1	203	9.20e-56	177.0	65.60	126.725562	2RGX_A	2RGX_A
59	214	1	203	2.75e-55	176.0	66.51	125.630579	4JLO_A	4JLO_A
60	214	6	209	5.53e-54	173.0	63.72	122.629407	1ZAK_A	1ZAK_A
61	185	8	190	4.30e-50	164.0	64.55	113.670640	1ZD8_A	1ZD8_A
62	185	7	189	5.91e-50	163.0	65.41	113.352609	2AK3_A	2AK3_A
63	213	6	213	2.11e-46	154.0	62.39	105.172226	4NTZ_A	4NTZ_A
64	182	28	207	7.44e-45	150.0	64.13	101.609458	2AR7_A	2AR7_A
65	182	6	185	6.22e-44	148.0	63.59	99.485974	3NDP_A	3NDP_A
66	182	1	155	1.03e-38	133.0	56.99	87.468675	1P4S_A	1P4S_A
67	182	21	175	1.86e-38	133.0	56.99	86.877657	2CDN_A	2CDN_A
68	209	1	218	3.97e-31	115.0	54.26	70.001372	3LOP_A	3LOP_A
69	205	13	184	4.85e-31	114.0	52.45	69.801159	5X6L_A	5X6L_A
70	209	1	218	5.24e-31	114.0	54.26	69.723816	2XB4_A	2XB4_A
71	205	11	182	5.83e-31	113.0	52.45	69.617121	5XRU_A	5XRU_A
72	205	11	182	6.84e-31	113.0	52.45	69.457350	5YCC_A	5YCC_A
73	205	13	184	8.11e-31	113.0	52.45	69.287040	5X6K_A	5X6K_A
74	213	13	192	9.57e-31	113.0	52.83	69.121505	5XZ2_A	5XZ2_A
75	213	11	190	9.75e-31	113.0	51.89	69.102871	5YCF_A	5YCF_A
76	205	11	182	1.05e-30	113.0	52.45	69.028763	5YCB_A	5YCB_A
77	182	11	164	2.79e-30	112.0	54.70	68.051511	5YCD_A	5YCD_A
78	184	12	167	3.34e-30	111.0	54.10	67.871582	3ADK_A	3ADK_A
79	185	32	188	1.76e-29	110.0	56.45	66.209654	3UMF_A	3UMF_A
80	184	12	167	2.07e-29	109.0	54.64	66.047419	1Z83_A	1Z83_A
81	184	16	171	2.46e-29	109.0	54.64	65.874806	7X7S_A	7X7S_A

82	214	7	185	9.98e-29	107.0	50.94	64.474385	3CM0_A	3CM0_A
83	184	11	166	1.32e-28	107.0	56.28	64.194751	7DE3_A	7DE3_A
84	184	11	166	1.65e-28	107.0	54.10	63.971607	8X1G_A	8X1G_A
85	168	1239	1418	1.72e-25	105.0	53.33	57.022303	7N6G_6	7N6G_6M
86	210	18	196	1.08e-24	97.8	53.55	55.185081	1UKY_A	1UKY_A
87	213	6	192	7.64e-24	95.5	49.77	53.228645	1TEV_A	1TEV_A
88	213	24	210	1.19e-23	95.5	49.77	52.785504	7E9V_A	7E9V_A
89	187	15	173	7.90e-22	90.1	49.47	48.590009	2BWJ_A	2BWJ_A
90	213	9	189	3.07e-20	85.9	47.44	44.930024	1QF9_A	1QF9_A
91	121	1411	1534	2.08e-08	55.5	50.00	17.688313	9FQR_x	9FQR_Xf
93	93	368	491	6.69e-07	50.8	46.40	14.217482	9FQR_x	9FQR_Xc
94	93	368	490	3.36e-06	48.9	45.16	12.603570	9D2F_D	9D2F_D
95	207	760	922	1.43e-05	47.0	42.01	11.155251	7N6G_6	7N6G_6A
96	126	976	1150	6.00e-03	39.3	42.29	5.115996	9D2F_A	9D2F_A
92	121	991	1160	2.30e-02	37.4	37.65	3.772261	9FQR_x	9FQR_Xf

The “top hits” are in the `hits` object. now we can download these to our computer. Put these in a wee sub-folder (directory) called “pdb” and use `gzip` to speed things up

```
#Download related PDB files
files <- get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/9R6U.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/9R71.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

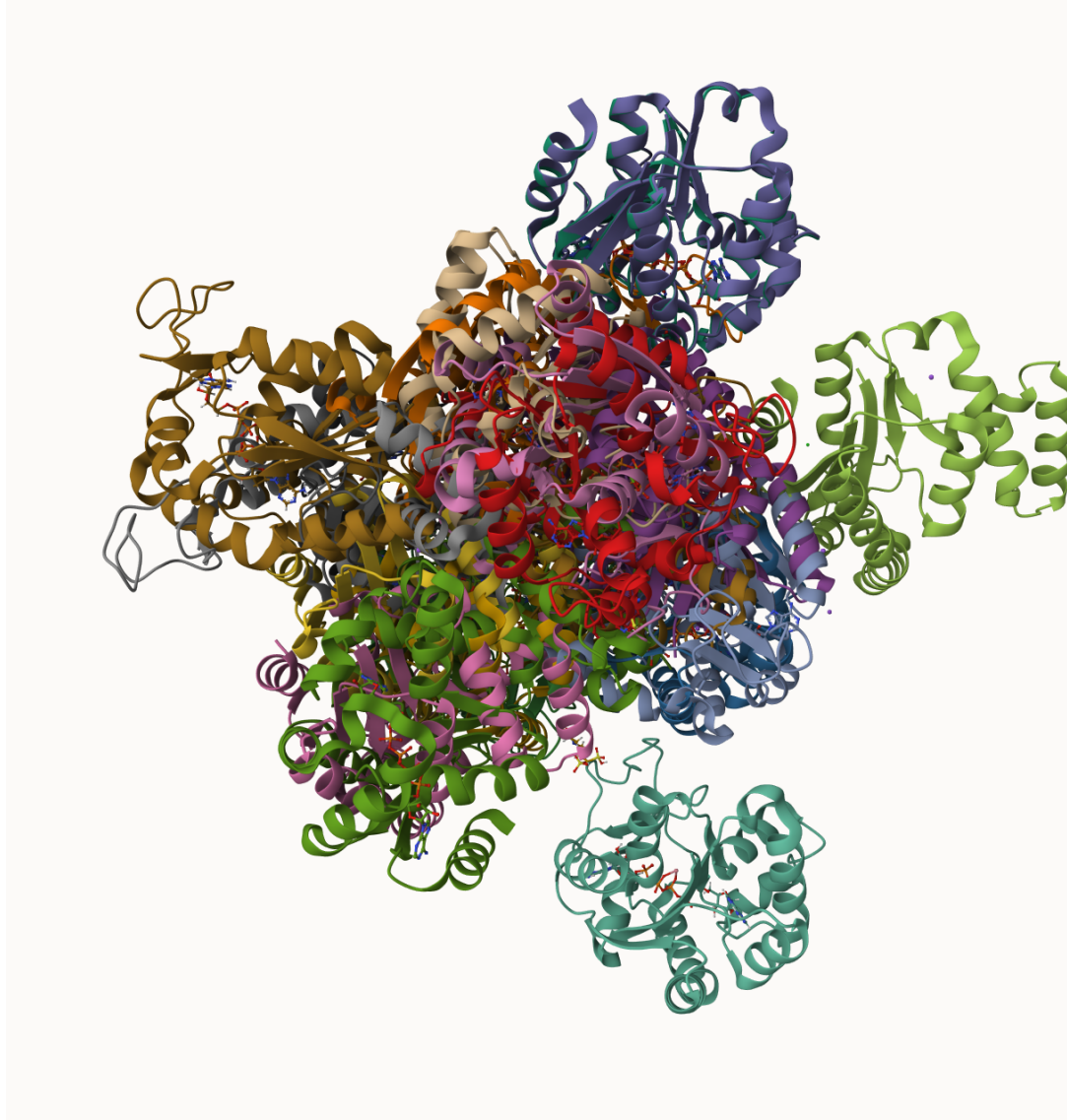
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download

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|
|=====| 100%

```



These look like a hot mess.

Next we will use the `pdbaln()` function to align and also optionally fit (i.e. superpose) the identified PDB structures.

This requires a BioConductor package called “msa” that we need to install. First we install BiocManager. Then we use `BiocManager::install("msa")`

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/8BQF_A.pdb
pdbs/split_chain/4X8M_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/9R6U_A.pdb
pdbs/split_chain/9R71_A.pdb
pdbs/split_chain/8Q2B_A.pdb
pdbs/split_chain/8RJ9_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/4X8H_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..
```

Extracting sequences

```
pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
```

PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 2 name: pdbs/split_chain/8BQF_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 3 name: pdbs/split_chain/4X8M_A.pdb
 pdb/seq: 4 name: pdbs/split_chain/6S36_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 5 name: pdbs/split_chain/9R6U_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 6 name: pdbs/split_chain/9R71_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 7 name: pdbs/split_chain/8Q2B_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 8 name: pdbs/split_chain/8RJ9_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 9 name: pdbs/split_chain/6RZE_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 10 name: pdbs/split_chain/4X8H_A.pdb
 pdb/seq: 11 name: pdbs/split_chain/3HPR_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 12 name: pdbs/split_chain/1E4V_A.pdb
 pdb/seq: 13 name: pdbs/split_chain/5EJE_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 14 name: pdbs/split_chain/1E4Y_A.pdb
 pdb/seq: 15 name: pdbs/split_chain/3X2S_A.pdb
 pdb/seq: 16 name: pdbs/split_chain/6HAP_A.pdb
 pdb/seq: 17 name: pdbs/split_chain/6HAM_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 18 name: pdbs/split_chain/8PVW_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 19 name: pdbs/split_chain/4K46_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

Have a wee peak at this new “alignment object” pdbs

pdbs

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				

```

[Truncated_Name:5] 9R6U_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:6] 9R71_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:7] 8Q2B_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:8] 8RJ9_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:9] 6RZE_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:10] 4X8H_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:11] 3HPR_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:12] 1E4V_A.pdb --MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:13] 5EJE_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:14] 1E4Y_A.pdb --MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:15] 3X2S_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:16] 6HAP_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:17] 6HAM_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:18] 8PVW_A.pdb --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:19] 4K46_A.pdb --MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMRLAA
[Truncated_Name:20] 4NP6_A.pdb NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMRLAA
*****  ***** *~*****
1 . . . 40

41 . . . 80
[Truncated_Name:1] 1AKE_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:2] 8BQF_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---
[Truncated_Name:3] 4X8M_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:4] 6S36_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:5] 9R6U_A.pdb VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:6] 9R71_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:7] 8Q2B_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:8] 8RJ9_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:9] 6RZE_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:10] 4X8H_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:11] 3HPR_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:12] 1E4V_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:13] 5EJE_A.pdb VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR
[Truncated_Name:14] 1E4Y_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:15] 3X2S_A.pdb VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR
[Truncated_Name:16] 6HAP_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR
[Truncated_Name:17] 6HAM_A.pdb IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR
[Truncated_Name:18] 8PVW_A.pdb VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:19] 4K46_A.pdb IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
[Truncated_Name:20] 4NP6_A.pdb IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE
~* *~*** *** ~^* **~*~^~^*~^~*~*
41 . . . 80

```

	81	.	.	.	120
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR	TIPQADAMKEAGINVDYVIEFDVPDELIVD			
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDALIVD			
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR	TIPQADAMKEAGINVDYVLEFDVPDELIVD			
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR	TIPQADGLKEVGVVVDYVIEFDVADSVIVE			
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR	TIPQADGLKEMGINVDYVIEFDVADDVIVE			
	****	*****~^** *~ ****~***** * ^**^			

81	.	.	.	120
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	121	.	.	.	160
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKA			
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDGTGEELTTRKD			
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAP	SGRVYHVKFNPPKVEGKDDVTGEELTTRKD			
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GET	SGRV-----D			

[Truncated_Name:19] 4K46_A.pdb	RMAGRRRAHLASGRITYHNVNPPKVEGKDDVTGEDLVIRE	
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRRAHLPSGRITYHVVNPPKVEGKDDVTGEDLVIRE	
	^ * ***	
	121 . . .	160
	161 . . .	200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG	
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG	
	* * * * * ^ * * * * * * * * * *	
	161 . . .	200
	201 . 216	
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-	
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:4] 6S36_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:5] 9R6U_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:6] 9R71_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:7] 8Q2B_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:8] 8RJ9_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:9] 6RZE_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:10] 4X8H_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:11] 3HPR_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:12] 1E4V_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:13] 5EJE_A.pdb	TKPVAEVRADLEKILG	

```

[Truncated_Name:14] 1E4Y_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb    TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb    TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb    TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb    TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

We could view these i R with **bio3dview** `view.pdbs()` function

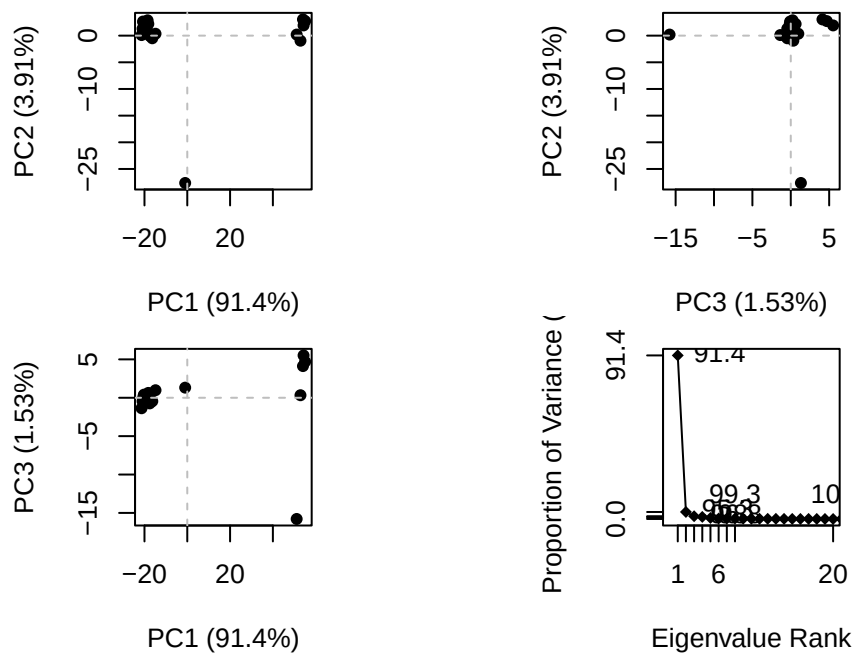
```
# library(bio3dview)

# view.pdbs(pdbs, colorScheme = "residue")
```

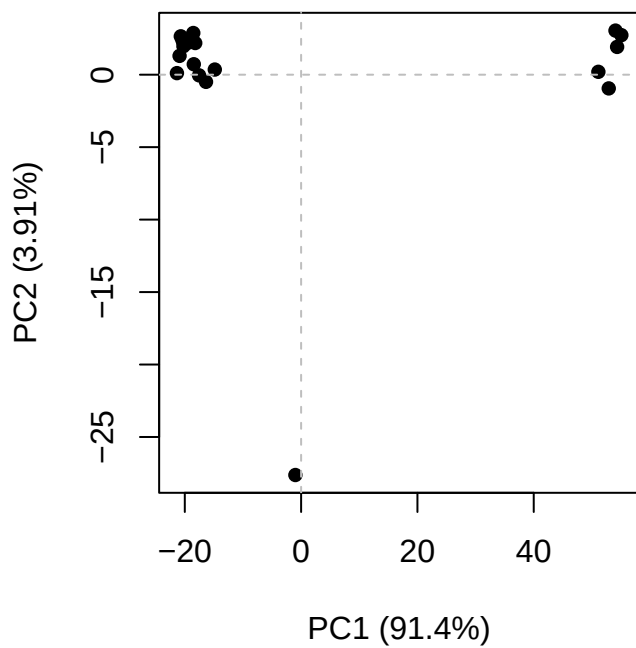
PCA

We can ru PCA on our `pdbs` object using the `pca()` function from **bio3d**:

```
pc.xray <- pca(pdbs)
plot(pc.xray)
```



```
plot(pc.xray, 1:2)
```



We can a visualization of the major conformational difference (i.e. large scale structure change) captured by our PCA analysis with the `mktrj()` function.


```
pc1 <- mktrj(pc.xray, file="pca.pdb")
```

Let's see in Mol-star: